

Homework #3

Shayan Karmaly
skarmaly3@gatech.edu

1 QUESTION 1

1.1 EAFC 26 – A Game of Slips and Mistakes

A game that I am familiar with is a video game called FC 26. Every year, EA comes out with the same game with similar modes and controls allowing users to predict a similar level of enjoyment in terms of functionality and overall satisfaction with the game. However, something always feels different despite it being the same idea of the game every year.

1.2 EAFC 26 Slip: “I Swear I Meant To Pass It!”

A slip that a player of the game might make is shooting the ball instead of passing it. A player’s intention is to pass the ball to a teammate, but they end up shooting the ball super far away towards the goal. A user might make this slip because they are under stress or pressure of the game, so their cognitive load is directed more towards the screen on which the game is being played than the task of using the controller. This makes the user more prone to performing this slip as they have less motor recruitment for their hands, forcing them to rely on their System 1 thinking, automatic and intuitive, whereas their System 2 thinking, calculated and quantitative, is focused on processing the game visually (Kahneman, 2013).

A way the interface could be changed to prevent that slip in the future is by mapping the controls better for the user to represent the relationship between the buttons to make actions better. The default settings might not be intuitive to some people, so EA has different button layouts for different people. However, the menu isn’t easily discoverable and wouldn’t be found by novices. I would suggest having an on-screen control overlay or contextual reminder before a user performs certain actions. For example, before a user needs to press *B* to shoot on goal, the system reminds them on screen and non-invasive to press it to shoot and score.

1.3 EAFC 26 Mistake: How To Power Shot... No Wonder You Can't Score

A mistake a player of the game might make is trying to perform a power shot without knowing the right button combination to do so. This mistake occurs in which the player's goal, to perform a power shot is correct, but they don't know how to perform it.

The player might make this mistake, especially novices, because they don't know how to. They'll see other players use it against them and will try to do so by pressing the right button to shoot, but it won't come out as intended.

A way the interface could be changed to prevent this mistake is by improving the discoverability of performing types of shooting techniques. Before shooting, the system can recognize that a player is about to shoot and display different techniques for shooting. For example, an on-screen, non-invasive display of the button combination for the power shot might make it easier for the user to perform the action and score using the power shot.

1.4 EAFC 26 Fut Champ Sweats

Something that makes the game challenging is not a slip or a mistake is getting ranked Gold III in Fut Champs. Fut Champs is a weekly tournament where you have fifteen games in total to play against random opponents who have also qualified for the tournament. To qualify for the tournament, you must participate in a qualifying tournament of 3 matches and win the whole thing without losing. Once you do qualify, you must wait for the weekend and complete the fifteen matches and win eight out of seven before the weekend ends to place in the gold division. Placing in Gold III guarantees you the best rewards for time and effort¹ in the number of games needed to play and win. It's easier said than done when your random opponents are usually professional e-sports players, YouTubers, experienced veterans, or overall experienced players. Scoring, let alone winning becomes extremely difficult the more challenging a player's opponent is purely do the skill of the player base.

¹ At least historically...

2 QUESTION 2

2.1 High-Load vs Low-Load For Muscle... Who Will Win?

As a passionate software engineer and bodybuilder I chose the paper “[Effects of Low- vs. High-Load Resistance Training on Muscle Strength and Hypertrophy in Well-Trained Men](#)” that compares the effects of low versus high load resistance training on muscular adaptations in well trained subjects (Schoenfeld et al., 2015).

2.2 Study Details

The study’s **null hypothesis** is that there would be no difference in strength, hypertrophy, or muscular endurance in comparing high-load and low-load resistance training. The study’s **alternative hypothesis** is that the high-load condition would have greater effects on strength and hypertrophy and that the low-load condition would have a superior impact on local muscle endurance. The **independent** variable in this study are the training types which are high and low load. The independent variables are of nominal type because they are categories of training. The **dependent** variables in this study are the change in strength, hypertrophy, and muscular endurance. These variables are of type ratio because they are measured numerically and compared.

2.3 ANOVA Usage

The statistical test used in this paper was ANOVA to compare the results of high and low-load training groups among well-trained individuals. It tested if the average of the results of these different groups in each category is different enough from each other. The difference or effect the researchers were looking for was what the difference in strength, hypertrophy, and muscle endurance would be in high-load training individuals compared to low-load training individuals.

They found that the well-trained individuals doing high-load training gained more strength while individuals doing low-load training gained more muscular endurance, but that both groups induced similar levels of hypertrophy (muscle gain).

I believe ANOVA was an appropriate statistical test because it allowed comparing the mean outcome across different categories. The conditions necessary for ANOVA are having an independent variable category type, interval or ratio

dependent variables, independent observations, and normal distribution of the dependent variables. This study meets these conditions with the independent variable being the category of high and low load training, the dependent variables of strength, hypertrophy, and muscular endurance being measured numerically, participants being assigned to only one group or the other, and measuring the dependent variables to determine whether high and low training effects them significantly or not.

3 QUESTION 3

3.1 Cronometer - The Life Saver

An interface from my everyday life that I would argue uses a good representation of its underlying content is my macro-tracking app *Cronometer*. The underlying content consists of my daily set macro and micro-nutrient targets that make up the total energy and food I consume in a day that aligns with my physique goals. That is connected to its representation via the *Targets* section that displays color coded progress bars for my total calories, protein, carbs, and fats for the day. If I swipe or tap the left arrow, I am presented with an *Energy Summary* view that displays my total energy consumed, expenditure, and how much of a deficit I'm in through color coded completion rings. If I swipe to the right most view, I can see my highlight nutrients which display all the micro-nutrients I'm getting in with a visual progress bar. These visual progress bars that act as representations for my daily intake fill proportionally as I log or remove food creating a direct mapping to the data they are displaying.

This representation exemplifies multiple instances of good representation. First, it succeeds at **making relationships explicit** which is shown between the relationship between how many calories are consumed and my daily targets that are represented with the visual progress bars and rings that give me real-time feedback of their current state. This provides an abstraction for users not having to interpret the numbers and they can visually assess their distance to their targets instead. This representation also **exposes natural constraints** by visually indicating the distance in approaching or exceeding a user's targets. This allows me to make decisions based on how many available nutrients I am able to eat in a day to achieve my goals.

3.2 The Conventional Oven – I Just Use My Air Fryer Now

I would argue my conventional oven does not use a good representation of its underlying content. The underlying content consists of the oven's current temperature, its heating status, and if it's reached the desired temperature. The mismatch between the representation and the underlying content occurs when the representation of this information isn't displayed in a way that gives useful feedback to the user that their action of the underlying content is being performed or not by the oven. Once I set the temperature, it displays three ticks ' - - - ' that doesn't indicate whether the oven is actively heating, when it will be ready, or how close it will be to ready. As a result of this poor feedback, I need to actively watch the screen to see if the ticks change into my desired temperature or not.

This representation violates several of the criteria that makes up a good representation. First, it fails to **make relationships explicit** because there is no relationship between the status of the temperature of the oven and what was set by the user. Without feedback on the progress or current state of the system, I have a hard time understanding the oven's current state therefore making things dependent on that knowledge, such as when to put my food in to cook, difficult. Second, it fails at **excluding extraneous details while highlighting relevant ones** because while it does include many of the relevant cooking settings such as bake, broil and timers, it doesn't indicate one of the most crucial details which is if the temperature has reached the desired temperature or not. Instead of emphasizing the representation of the oven's status and temperature, the interface displays less relevant ones making it difficult for the user to understand the status of the oven and make decisions quickly and effectively.

4 QUESTION 4

4.1 Cognitive Task Analysis of GATech Email Login

Step 1: Navigating to the GT Login Page

Here the user opens a browser to login to their email. The knowledge required is that a user needs to know how to open a browser and use the correct URL. The source of this knowledge comes from previous general computer knowledge and from memory or a saved bookmark.

Step 2: Login into GT Account and Password

Users need to know their unique details which they have setup already. The interface prompts the user to input it via the text boxes. If the user has forgotten these details, the interface provides an option to retrieve those details using links at the bottom of the page. The source of knowledge is their memory or saved in a keychain or written down on a piece of paper.

Step 3: Enter Authentication Code in Duo Mobile

After submitting valid credentials, users are prompted by entering the code they see on the browser in their Duo mobile app. The knowledge required here is that a second authentication of Duo is required and that it is part of the login sequence. The source of this knowledge comes from previously doing this beforehand as well as the interface guiding the user by showing an on-screen guide/prompt to check their Duo app.

Step 4: Verify device on the Duo app

After seeing the on-screen instructions, users need to open their Duo app and input the same code as seen on the browser in their app as well. The knowledge required here is being able to unlock their phone, locate their Duo app, and know that Duo will prompt them to input a code. The source of knowledge comes from general knowledge of using a smartphone, remembering where their app is, and on-screen instructions that show up when they open the mobile app when Duo recognizes it needs a code from the user. The user then needs to know to wait for the code prompting to pop up which only comes from previous usage of the app. Once it does show up, Duo guides the user over the bridge of execution by displaying a keypad and a spot for three digits to enter.

Step 5: Submitting the Code

Once a user submits the code, a bright green screen presents itself guiding the user over the gulf of evaluation that the authentication has been achieved. No prior knowledge is needed here as the interface makes it obvious that they've been authenticated.

Step 4: Trust GaTech.edu

Once the user is back on their browser and they've been authenticated, they're prompted to trust GaTech.edu. Prior knowledge is required here to know their

institution and to be able to trust them. Once they click *continue*, they are presented with their email landing page.

4.2 3 Hueristics of GATech Email Login

1. Visibility of System Status

This process follows this heuristic well because after entering a username and password, the system immediately presents feedback that it needs Duo authentication from the user. Once the user accepts this, they are immediately shown to another screen that asks them to input the code on their other interface, the Duo mobile app. After entering the code, the app shows them instant feedback via a green screen signifying valid authentication and to head back to their browser. On their browser, the screen immediately reflects that they've been verified and are heading to their institutions page. At any step of the process if a user inputs incorrectly, the system gives them feedback by either displaying red text outlining what they did incorrectly or a screen with what they did incorrectly and how to fix it.

2. Match Between System and the Real World

This process follows this heuristic well because this process is a similar two-factor authentication process to other apps such as Google or Microsoft Authenticator in which users can rely on prior knowledge of the real world and it's applications. The interface also uses easily understandable language for the user to indicate this is an extra authentication process for security reasons making it easy for them to think about the task and not the interface itself.

3. Error Prevention and Recovery

This process follows this heuristic well because this system prevents errors from occurring and makes it easy if errors do occur. If a user inputs the wrong GT credentials, they can reset their password at the bottom of the page. If they input the wrong Duo code, the Duo app presents them with an *invalid verification code* page with an error message and useful tips on how to log in. If they input the wrong code, the browser also immediately provides feedback with an *invalid verification code* page and prompts them to either try again or use *other options* to sign in in case the previous one didn't work well for the user.

5 REFERENCES

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